

Initial prompt to Claude Code:

In the folder EIC-LHeC, there are files WEV1301 (tex and pdf), that describe vibe-coding for EIC hybrid version. The Table 2 shows the parameters of electron and proton beams. I would like to ask you to design final doublets (FD) for both electron and proton beams (these two designs are separate, but done in the same way, you would just need to change the input parameters). The relevant input parameters are the beam energy, the type of the particle, and the horizontal and vertical beam divergence at the IP. Let's assume that the L^* is the distance from the IP (Interaction Point), to the beginning of the first quadrupole in the FD. The FD will consist of two quadrupoles of different polarity. I would like to ask you to write a code, that will trace the beam, coming from the IP, go through the FD, and turn into a parallel beam. For tracing the beam, please use four trajectories with zero position at the IP and initial angles equal to plus and minus beam divergence from the table, for horizontal and vertical planes. Assume, initially, that L^* is equal to 2m, the length of each quadrupole of the FD is also 2m, and distance between them is 0.5m. Create a picture of X and Y beam envelope coming from the IP and exiting the FD (as the parallel beam, for which you will need to adjust the strength of the quadrupoles). Please make one picture for X and Y beam envelope for electron beam, and another picture for X and Y beam envelope for the proton beam (display them, and also write to PNG files – I will later plan to use one of the pictures in this paper). The choice of the polarity of the quadrupoles in the FD needs to be done in such a way that the maximum beam size in the FD will be minimized. Please also print the maximum beam size in the FD on the pictures. Please let me know if you have additional questions at this moment.

Response from Claude Code, who created the code fd_design.py and pushed the code and pictures to GitHub:

Everything is pushed. Here's a summary of the results:

Electron beam ($E = 10$ GeV):

- Best polarity: QD–QF (Q1 focuses vertically first, because $\theta_y = 149.9 \mu\text{rad} > \theta_x = 30.6 \mu\text{rad}$)
- $k_1 = -0.3190 \text{ m}^{-2}$, $|G_1| = 10.6 \text{ T/m}$
- $k_2 = +0.1671 \text{ m}^{-2}$, $|G_2| = 5.6 \text{ T/m}$ ($B\rho = 33.4 \text{ T}\cdot\text{m}$)
- Max beam size: 0.40 mm (vertical)

Proton beam ($E = 275$ GeV):

- Best polarity: QF–QD (Q1 focuses horizontally first, because $\theta_x = 306.3 \mu\text{rad} > \theta_y = 130.6 \mu\text{rad}$)
- $k_1 = +0.3190 \text{ m}^{-2}$, $|G_1| = 292.6 \text{ T/m}$
- $k_2 = -0.1671 \text{ m}^{-2}$, $|G_2| = 153.3 \text{ T/m}$ ($B\rho = 917.3 \text{ T}\cdot\text{m}$)
- Max beam size: 1.43 mm (vertical)

Key observations:

1. The same k-value magnitudes solve both beams (only the polarity flips) — this is because the geometry (L^* , L_q , gap) is identical. The divergence asymmetry determines which plane Q1 focuses first.
2. The proton gradients (293 T/m, 153 T/m) are very high and would require superconducting quadrupoles (likely Nb₃Sn for the inner quad), which is consistent with the IR magnets discussion in the paper.
3. The electron gradients (10.6 T/m, 5.6 T/m) are modest and achievable with conventional Nb-Ti SC or even normal-conducting magnets.

✧ Crunched for 18m 41s

☒ Background command "Run FD design code (fixed)" completed (exit code 0)

☒ The background task completed successfully (exit code 0). Both figures are already saved and pushed to GitHub.

Second prompt to Claude Code:

This is wonderful, great design, thank you! Let's now analyze what magnet technology we need to use for the designed Final Doublets. Let's assume that the aperture of the quadrupoles is round and that two quadrupoles in each FD use the same technology. Let's assume that the value of magnetic field at the inner aperture of superconducting quadrupole (so called bore field) can reach the following levels, depending on magnet technology and temperature: a) NbTi quadrupole at 1.9 K temperature: 7 Tesla aperture field; b) NbTi quadrupole at 4.2 K temperature: 8.5 Tesla aperture field; c) Nb₃Sn quadrupole at 4.2 K temperature: 12 Tesla aperture field. Taking this into account, please evaluate the ratio of the maximum possible quadrupole aperture, for a particular technology, to the maximum beam size that you calculated in the previous step. Calculate it for each plane and define it as N_X and N_Y . Please output the results in terms of a table, or a list, for each of the technology.

Here are the results, organized by beam:

Magnet Technology Analysis for Final Doublets

Formula: $R_{\max} = B_{\text{bore}} / G$ (maximum aperture radius for a given technology at gradient G), then $N_X = R_{\max} / \max_x$, $N_Y = R_{\max} / \max_y$

Proton FD (275 GeV) — $G(Q1) = 292.6 \text{ T/m}$, $G(Q2) = 153.3 \text{ T/m}$, $\max|x| = 0.818 \text{ mm}$, $\max|y| = 1.434 \text{ mm}$

Technology	R_max Q1 [mm]	N_X (Q1)	N_Y (Q1)	R_max Q2 [mm]	N_X (Q2)	N_Y (Q2)
NbTi, 1.9 K (7.0 T)	23.9	29.2	16.7	45.7	55.8	31.8
NbTi, 4.2 K (8.5 T)	29.0	35.5	20.3	55.4	67.8	38.7
Nb ₃ Sn, 4.2 K (12.0 T)	41.0	50.1	28.6	78.3	95.7	54.6

Electron FD (10 GeV) — $G(Q1) = 10.64 \text{ T/m}$, $G(Q2) = 5.58 \text{ T/m}$, $\max|x| = 0.336 \text{ mm}$, $\max|y| = 0.400 \text{ mm}$

Technology	R_max Q1 [mm]	N_X (Q1)	N_Y (Q1)	R_max Q2 [mm]	N_X (Q2)	N_Y (Q2)
NbTi, 1.9 K (7.0 T)	658	1958	1645	1255	3734	3136
NbTi, 4.2 K (8.5 T)	799	2378	1997	1523	4534	3808
Nb ₃ Sn, 4.2 K (12.0 T)	1128	3357	2820	2151	6400	5376

Key conclusions:

Proton FD — All three SC technologies are geometrically feasible ($N \gg 1$ in all cases). The most constrained case is Q1 in the vertical plane (smallest N). A minimum practical aperture of $\sim 10\text{--}15 \times$ beam sigma is generally required in collider IR designs to accommodate orbit offsets, halo, and tolerances. That threshold is comfortably cleared by all three technologies (minimum $N_Y = 16.7$ for NbTi at 1.9 K). However, the required gradient of **293 T/m** for Q1 mandates **superconducting technology** — normal-conducting quads (typically $\leq 100 \text{ T/m}$) are ruled out. Nb₃Sn at 4.2 K provides the most margin ($N_Y = 28.6$ for Q1), making it the preferred technology if aperture flexibility and operational margin are priorities. NbTi at 4.2 K (simpler cryogenics than 1.9 K) is also viable.

Electron FD — N values are in the thousands. The required gradients (10.6 and 5.6 T/m) are so low that **superconducting magnets are not needed at all** — normal-conducting room-temperature quadrupoles would work easily.

One note on the NbTi specifications: the numbers you provided show NbTi at 1.9 K (7 T) giving a *lower* bore field than at 4.2 K (8.5 T), which is the reverse of the usual trend (lower temperature → higher critical current → higher field). You may want to verify these values before using them in the paper — for typical LHC-style NbTi IR quads, 1.9 K operation achieves ~7–8.5 T bore, and 4.2 K would be somewhat less. If the two values are swapped, the conclusions are unchanged qualitatively.

Prompt: confirmed typo in the previous prompt

Claude Code Response:

Done — corrected results with NbTi 1.9 K → 8.5 T and NbTi 4.2 K → 7.0 T. The tables now show the proper ordering (1.9 K gives larger aperture margin than 4.2 K for NbTi, as expected physically).

Updated key numbers for the proton Q1 (most constrained):

Technology	R_max Q1	N_X	N_Y
NbTi, 1.9 K (8.5 T)	29.0 mm	35.5	20.3
NbTi, 4.2 K (7.0 T)	23.9 mm	29.2	16.7
Nb ₃ Sn, 4.2 K (12.0 T)	41.0 mm	50.1	28.6

The conclusions are unchanged: all three SC technologies are feasible for the proton FD (min N_Y = 16.7 > 10), the electron FD doesn't need SC magnets at all, and Nb₃Sn gives the most comfortable margin.